SASSCAL
Southern African Science Service Centre for Climate Change and Adaptive Land Management
WWW.SASSCAL.ORG

TIPPECC

With funding from the:
Federal Ministry of Research, Technology and Space

GERICS
Climate Service Center Germany

TIPPECC Summer School
on the Analysis of Climate Model Data

Introduction to the NetCDF

Dr. Torsten Weber & Franziska Zander

WITS Rural Facility
09 – 12 March 2026

University of the Witwatersrand, Johannesburg
Global Change Institute
The University of Zambia
University of Botswana
GOBABEB Namib Research Institute
Friedrich-Schiller-Universität Jena

Agenda

- NetCDF-format & software
- CF Convention
- View data text based & Practice
- Visualise data & Practice

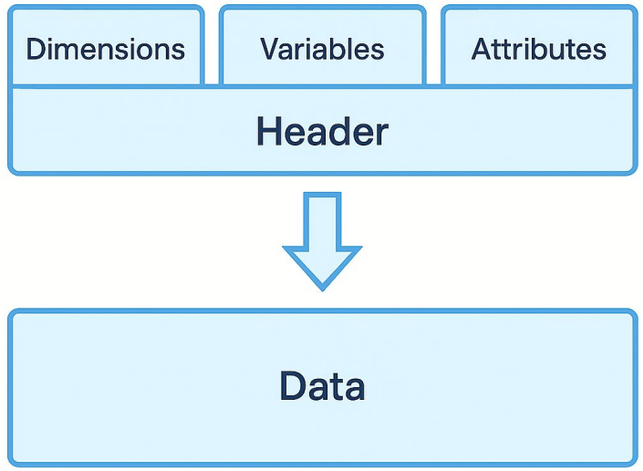
NetCDF-Format & Software

What is NetCDF?

- **NetCDF** stands for **Network Common Data Form**.
- scientific data format designed to store **multidimensional array data** together with **metadata**.
- standard format in climate, meteorology, and oceanography

- **NetCDF-Software**: Library to read, create, edit & validate NetCDF files
  - The NetCDF homepage can be found at http://www.unidata.ucar.edu/software/netcdf/.
  - The NetCDF source code is hosted on GitHub and is available at http://github.com/Unidata/netcdf-c. (Source: http://www.unidata.ucar.edu)

NetCDF File Structure



(created with MS Copilot)

CF conventions

- Climate and Forecast Metadata Conventions
- community standard that defines how climate and geoscientific **data** should be **described inside NetCDF files** (→ software can automatically understand)
  - **Standard Names** *(“air_temperature”)*
  - **Units** (*"K„*)
  - **Coordinates**
  - **Time Handling** (calendars, …*time:calendar* = *"360_day"*)
  - **Cell Methods** (*tas:cell_methods* = *"time: mean„)*
  - **Grids & Projections** (Regular lat/lon grids, rotated pole grids (CORDEX), ….)

```
float lat(lat)
  lat:standard_name = "latitude"
  lat:units = "degrees_north"

float lon(lon)
  lon:standard_name = "longitude"
  lon:units = "degrees_east"
```

CF conventions

- Major projects require CF compliance:
  - CMIP
  - CORDEX
  - ERA5 (mostly CF compliant)
  - ESGF datasets

  - …and TIPPECC ;-)

  - More information: https://cfconventions.org/
  - (Standard names: https://cfconventions.org/Data/cf-standard-names/current/build/cf-standard-name-table.html)

NetCDF-Format

How can I see the data description and the contents of a NetCDF file?

- ***ncdump*** text-based metadata and data inspection
- ***ncview*** graphical view of the data

```
>:ncdump --help
ncdump [-c|-h][-v ...][[-b|-f][c|f]][-l len][-n name][-p n[,n]][-k][-x][-s][-t|-i][-g
...][-w][-Ln]file

 [-c]                Coordinate variable data and header information
 [-h]                Header information only, no data
 [-v var1[,...]]     Data for variable(s) <var1>,... only
 [-b [c|f]]          Brief annotations for C or Fortran indices in data
 [-f [c|f]]          Full annotations for C or Fortran indices in data
 [-l len]            Line length maximum in data section (default 80)
 [-n name]           Name for netCDF (default derived from file name)
 [-p n[,n]]          Display floating-point values with less precision
 [-k]                Output kind of netCDF file
```

- continued on the next slide -

NetCDF-Format

```
 [-s]              Output special (virtual) attributes
 [-t]              Output time data as date-time strings
 [-i]              Output time data as date-time strings with ISO-8601 'T' separator
 [-g grp1[,...]]   Data and metadata for group(s) <grp1>,... only
 [-w]              With client-side caching of variables for DAP URLs
 [-x]              Output XML (NcML) instead of CDL
 [-Xp]             Unconditionally suppress output of the properties attribute
 [-Ln]             Set log level to n (>= 0); ignore if logging not enabled.
 file              Name of netCDF file (or URL if DAP access enabled)

netcdf library version 4.8.1 of Apr 25 2022 17:44:24 $
```

Most important information of a NetCDF-file is the header, which can be displayed with

`>:ncdump -h file`

The header of a NetCDF-file contains information of:

- dimensions of the variable(s)
- variables
- global attributes

NetCDF – Header Information

```
Example:
>:ncdump -h tas_AFR-44_MPI-M-MPI-ESM-LR_historical_r1i1p1_MPI-CSC-EMO2009_v1_mon.nc
netcdf tas_AFR-44_MPI-M-MPI-ESM-LR_historical_r1i1p1_MPI-CSC-REMO2009_v1_mon {
dimensions:
        x = 194 ;
        y = 201 ;
        nv4 = 4 ;
        time = UNLIMITED ; // (672 currently)
        bnds = 2 ;
variables:
        float lon(y, x)
                lon:standard_name = "longitude" ;
                lon:long_name = "longitude coordinate" ;
                lon:units = "degrees_east" ;
                lon:_CoordinateAxisType = "Lon" ;
                lon:bounds = "lon_bnds" ;
```

Variable

Variable Attributes

- continued on the next slide -

NetCDF – Header Information

```
float lon_bnds(y, x, nv4) ;
        float lat(y, x) ;
                lat:standard_name = "latitude" ;
                lat:long_name = "latitude coordinate" ;
                lat:units = "degrees_north" ;
                lat:_CoordinateAxisType = "Lat" ;
                lat:bounds = "lat_bnds" ;
        float lat_bnds(y, x, nv4) ;
        ……
        double time(time) ;
                time:standard_name = "time" ;
                time:long_name = "time" ;
                time:bounds = "time_bnds" ;
                time:units = "days since 1949-12-1 00:00:00" ;
                time:calendar = "proleptic_gregorian" ;
                time:axis = "T" ;
        double time_bnds(time, bnds) ;
```

Variable

Variable Attributes

Variable

Variable Attributes

- continued on the next slide -

NetCDF – Header Information

```
float tas(time, y, x)
```

Variable

```
tas:standard_name = "air_temperature" ;
tas:long_name = "Near-Surface Air Temperature" ;
tas:units = "K"
tas:coordinates = "height lat lon" ;
tas:_FillValue = 1.e+20f ;
tas:missing_value = 1.e+20f ;
tas:comment = "daily-mean near-surface
             (usually, 2 meter) air temperature." ;
tas:cell_methods = "time: mean" ;
tas:history = "2015-07-16T15:09:26Z altered by CMOR:
               Treated scalar dimension: \'height\'.
               2015-07-16T15:09:26Z altered by CMOR:
               replaced missing value flag (-1e+20)
               with standard missing value (1e+20)." ;
tas:associated_files = "gridspecFile:
                        gridspec_atmos_fx_MPI-CSC-
                        REMO2009_historical_r0i0p0.nc" ;
```

Variable Attributes

- continued on the next slide -

NetCDF – Header Information

```
// global attributes:
                :CDI = "Climate Data Interface version 1.7.0
                       (http://mpimet.mpg.de/cdi)" ;
                :history = "Thu Mar 03 09:50:44 2016: cdo mergetime
                 tas_AFR-44_MPI-M-MPI-ESM-LR_historical_r1i1p1_MPI-CSC-
                 REMO2009_v1_mon_195001-195012.nc tas_AFR-44_MPI-M-MPI-
                 ESM-LR_historical_r1i1p1_MPI-CSC-
                 REMO2009_v1_mon_195101-196012.nc…",
                :source = "MPI-CSC-REMO2009" ;
                :institution = "Helmholtz-Zentrum Geesthacht,
                 Climate Service Center, Max Planck Institute for
                 Meteorology" ;
                :Conventions = "CF-1.4" ;
                :institute_id = "MPI-CSC" ;
                :experiment_id = "historical" ;
                :model_id = "MPI-CSC-REMO2009" ;
                :contact = "claas.teichmann@hzg.de" ;
                :comment = "CORDEX Africa RCM REMO 0.44 deg AFR-44" ;
```

- continued on the next slide -

NetCDF – Header Information

```
// global attributes:
                :references = "http://www.remo-rcm.de/" ;
                :initialization_method = 1 ;
                :physics_version = 1 ;
                :tracking_id = "30fe24ea-5ff6-4bf1-b123-6d9b9a24367" ;
                :CORDEX_domain = "AFR-44" ;
                :driving_experiment = "MPI-M-MPI-ESM-LR, historical,
                 r1i1p1" ;
                :driving_model_id = "MPI-M-MPI-ESM-LR" ;
                :driving_model_ensemble_member = "r1i1p1" ;
                :driving_experiment_name = "historical" ;
                :rcm_version_id = "v1" ;
                :product = "output" ;
                :experiment = "historical" ;
                :frequency = "mon" ;
                :creation_date = "2015-07-16T15:09:26Z" ;
                :project_id = "CORDEX" ;
```

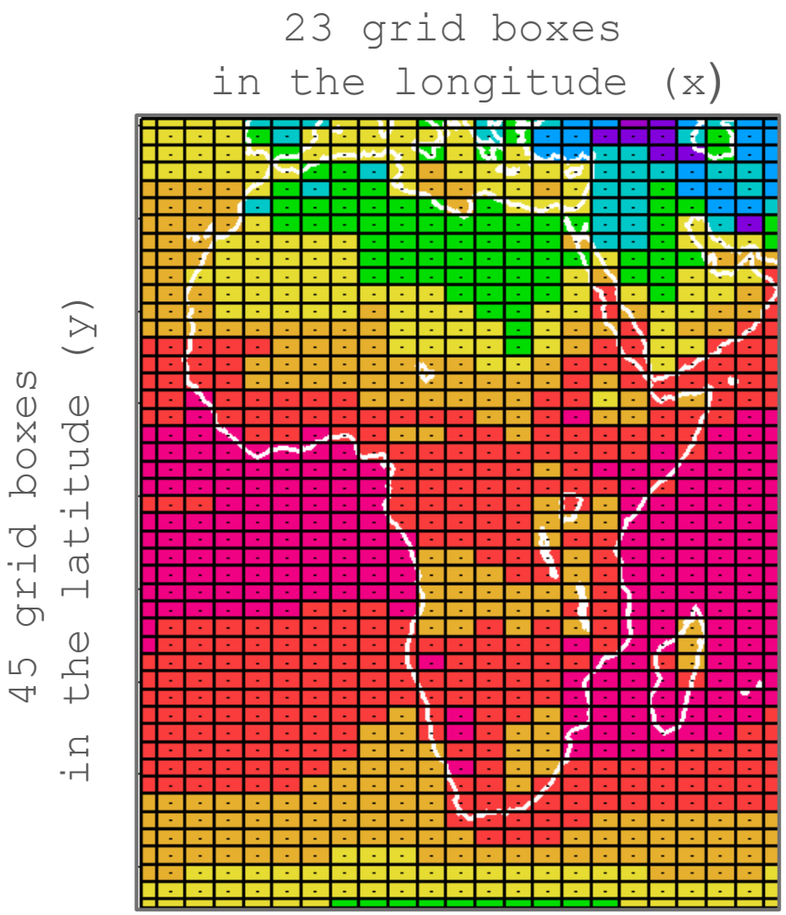
- continued on the next slide -

NetCDF – Header Information

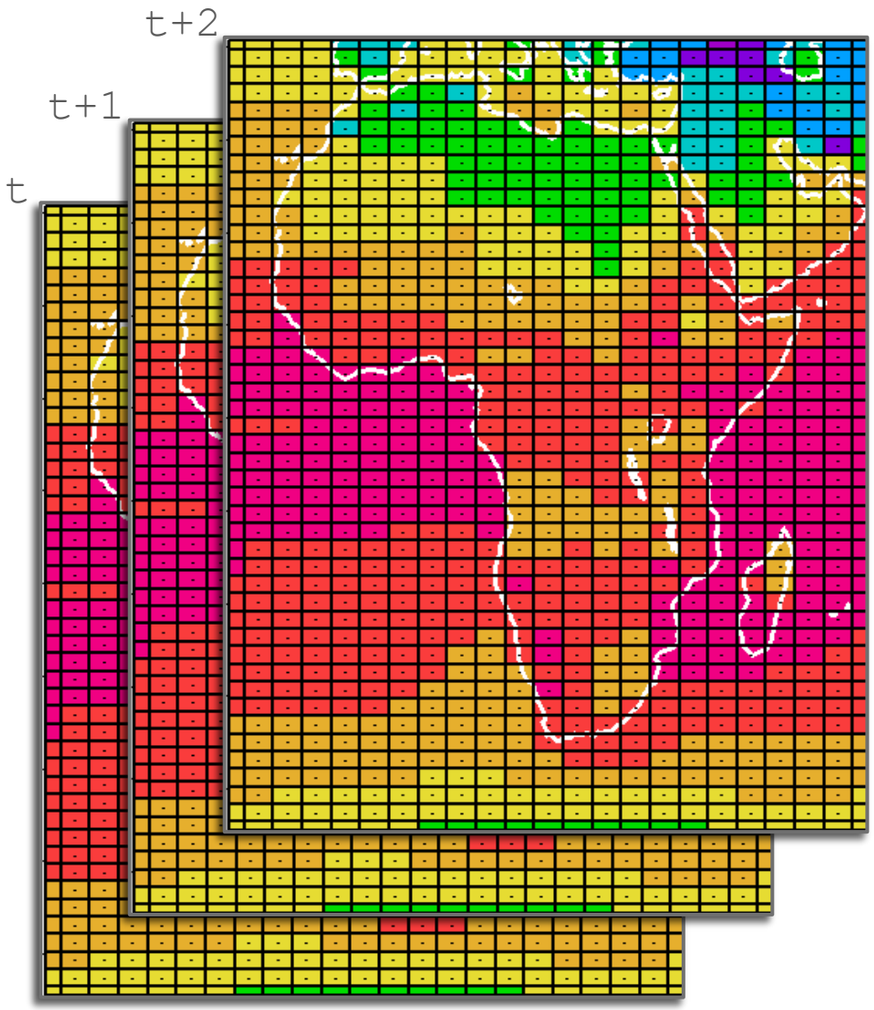
```
// global attributes:
                :table_id = "Table mon (Mar 2015)
 db0b230ff4a2c922671f0c21978b6242" ;
                :title = "MPI-CSC-REMO2009 model output prepared for CORDEX
 historical" ;
                :modeling_realm = "atmos" ;
                :realization = 1 ;
                :cmor_version = "2.9.1" ;
                :CDO = "Climate Data Operators version 1.7.0
(http://mpimet.mpg.de/cdo)" ;}
```

NetCDF – Dimensions

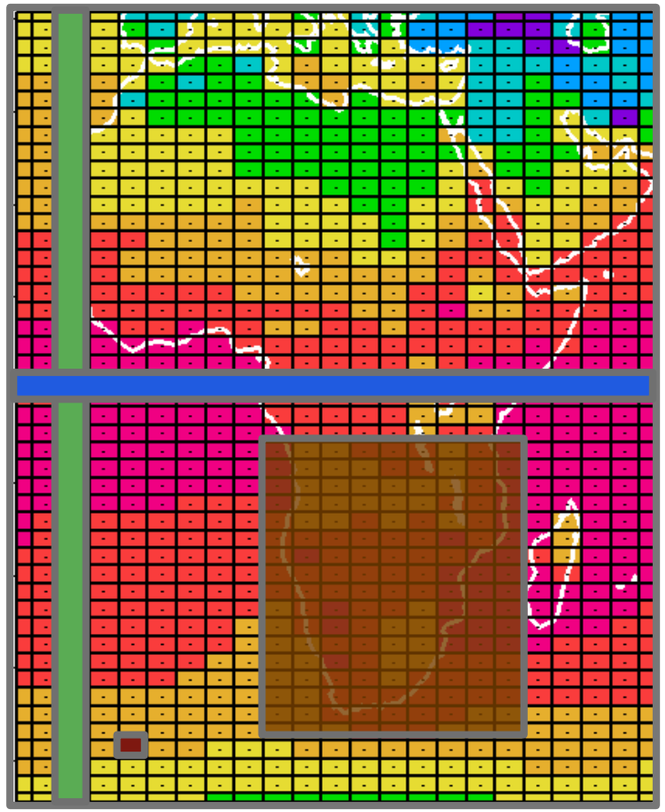
Two dimensional array (y,x)



Three dimensional array (t,y,x)



Cutting out regions



46 * 23 array –> 2d data
n * m array –> 2d regional selection
46 * 1 array –> 1d meridional selection
1 * 23 array –> 1d zonal selection
1 * 1 array –> single grid box
scalar –> field mean

CORDEX – Filename Convention

Variable

Earth System Model (GCM)

Ensemble Member

RCM Version ID

File Suffix: NetCDF

tas_AFR-22_MPI-M-MPI-ESM-LR_historical_r1i1p1_GERICS-REMO2015_v1_day_195001-195012.nc

Domain

Experiment

Regional Climate Model (RCM)

Frequency

Start Time - End Time

NetCDF – Header Information

Practice

1. Which unit and standard name does the precipitation variable (pr) in the file "Fut_TIPPECC_b_GERICS-REMO2015_v1_MPI-M-MPI-ESM-LR_pr_Afr_day_2081_2100.nc" have?

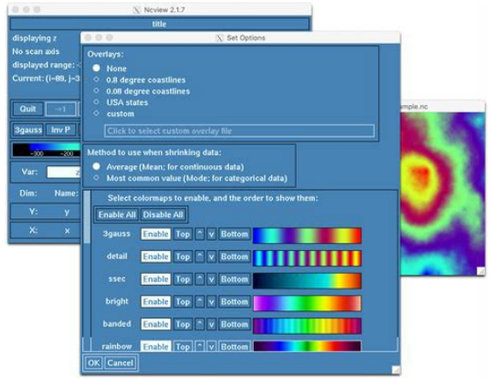
2. How many dimensions does the variable (pr) have?

3. When is the last time step ? (ncdump –v time –i …)

Notice: Change into the directory with `cd /home/user1/training/data/raw` and use the command `ncdump -h <file.nc>`

NetCDF quick data view

- **ncview**
  - **Best for:**
    - Fast inspection of model output
    - Clicking through time steps
    - Checking periodicity issues
    - Spotting missing values or artifacts
    - HPC login node quick checks

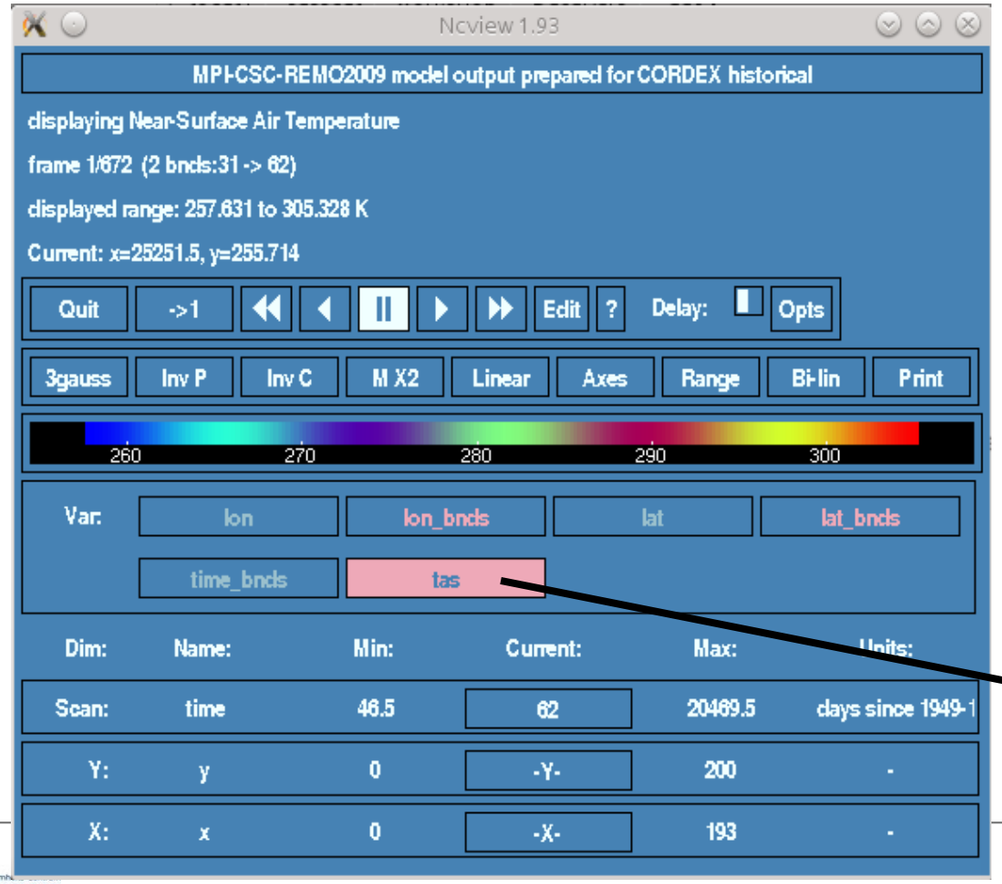


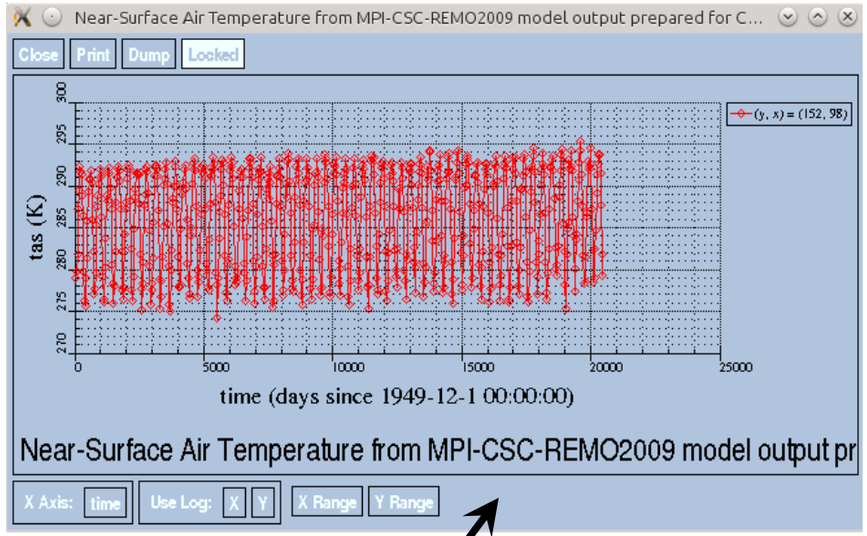
- **QGIS**
  - **Best for:**
    - Georeferenced datasets
    - Map projections
    - Overlaying shapefiles
    - Combining raster + vector data
    - Producing maps

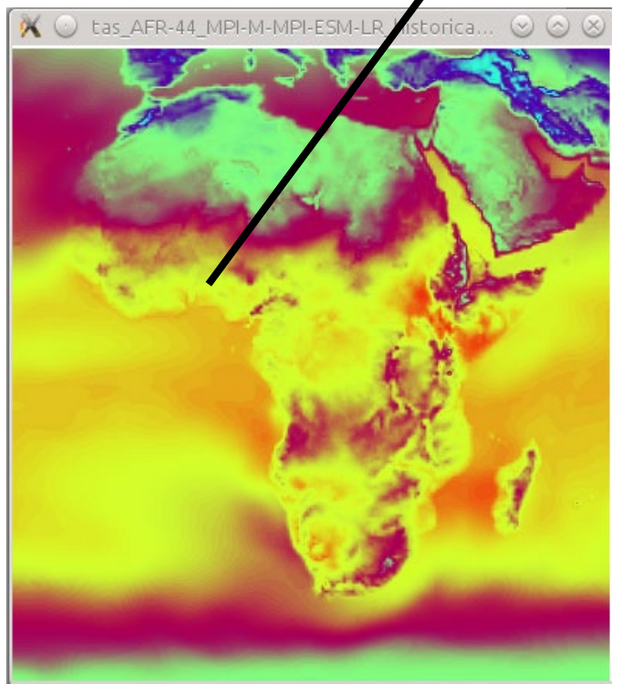
Quick data view

QGIS or ncview

```
>:ncview file.nc
```

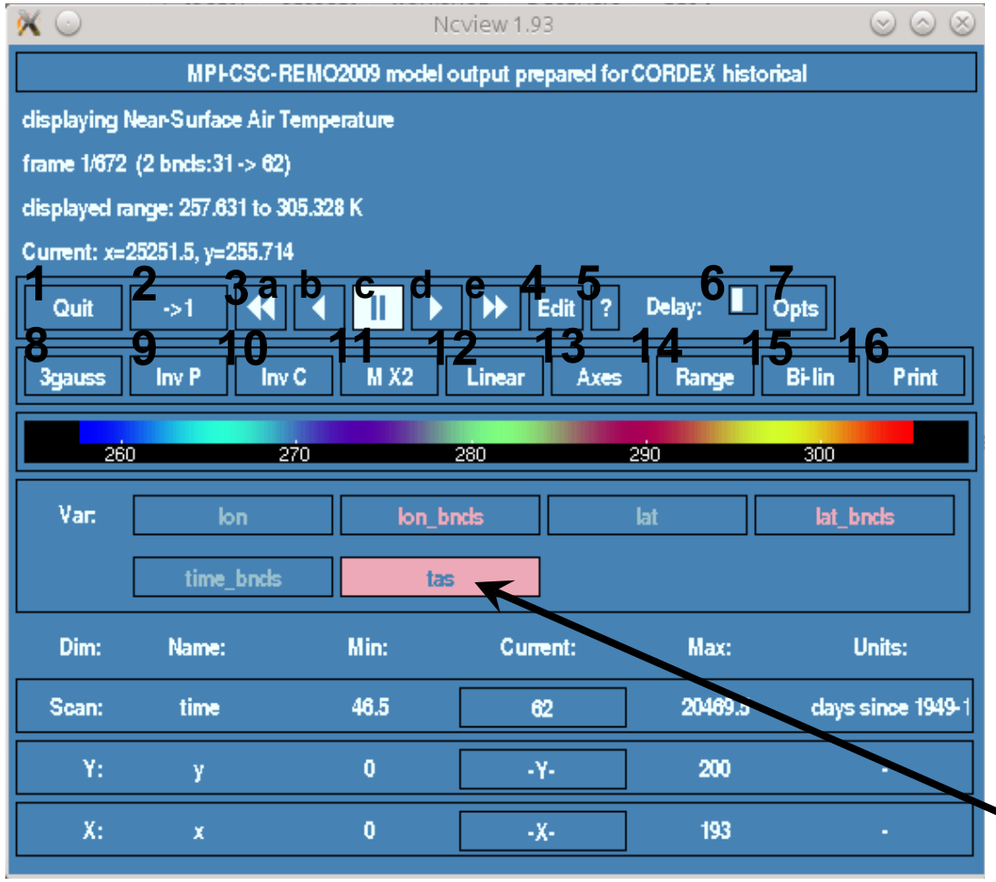






Quick data view

```
>: ncview file.nc
```



first row:
1: quit ncview
2: back to the first screen
3a-e: move through the data
4: edit data
5: information about the variable
6: delay between two screen
7: options for map overlay

second row:
8: colour tables
9: invert plot (turn upside down)
10: invert colours
11: magnify plot (increase: left, decrease right mouse button click)
12: colour filter (high, low, linear)
13: change axes
14: set data range (click with the right mouse button gives the best range of colours
15: bilinear interpolation or replicate pixels
16: print plot to postscript-file

third row:
choosen variable

Quick data view: practice

Practice

1. Display the variable “2tm” in the file “t2m_ERA5_global.2021_2025.nc”.

2. Use the corresponding button to move through the timeframes.

Notice: Change into the directory with `cd /home/user1/training/data/raw` and use the command `ncview <file.nc>`

Thank you for your attention!